

Differentiable pull-in solver

$$R(Y, \Lambda) = 0$$

$$J(Y, \Lambda) v = 0$$

$$\|v\| = 1$$

exact $\partial V / \partial d(x)$, no surrogate

Inverse design

22.86 $\rightarrow$ 13.60 V

−41%, shape alone

15 s, laptop CPU

no new mask

Amortized net

0.04 ms/design

spec $\rightarrow$ geometry

0 stored sims

0.56% spec error

Safe-operation RL

+12.8% travel

0.0% destroyed

learns its ceiling

not measurable

No training dataset. No commercial licence.

8 independent sources, 1967–2025 · gradient exact to 0.00%